REQUIREMENTS ENGINEERING

COMP6100001 - SOFTWARE ENGINEERING
SESSION 06

UNIVERSITAS **BINA NUSANTARA**

SUBJECT MATTER EXPERT

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SESSION 06 – REQUIREMENT ENGINEERING

LEARNING OBJECTIVES

At the end of this session, students will be expected to be able to:

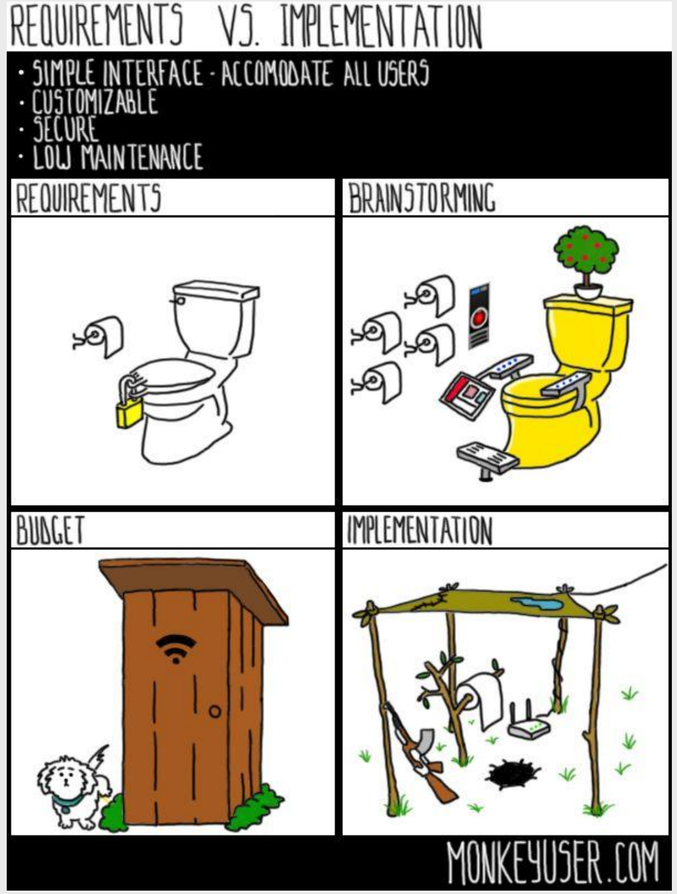
- Explain the role and importance of RE (including Agile)
- Identify and analyze stakeholders
- Select elicitation techniques
- Explain and apply the seven RE tasks
- Apply RE to a realistic case study

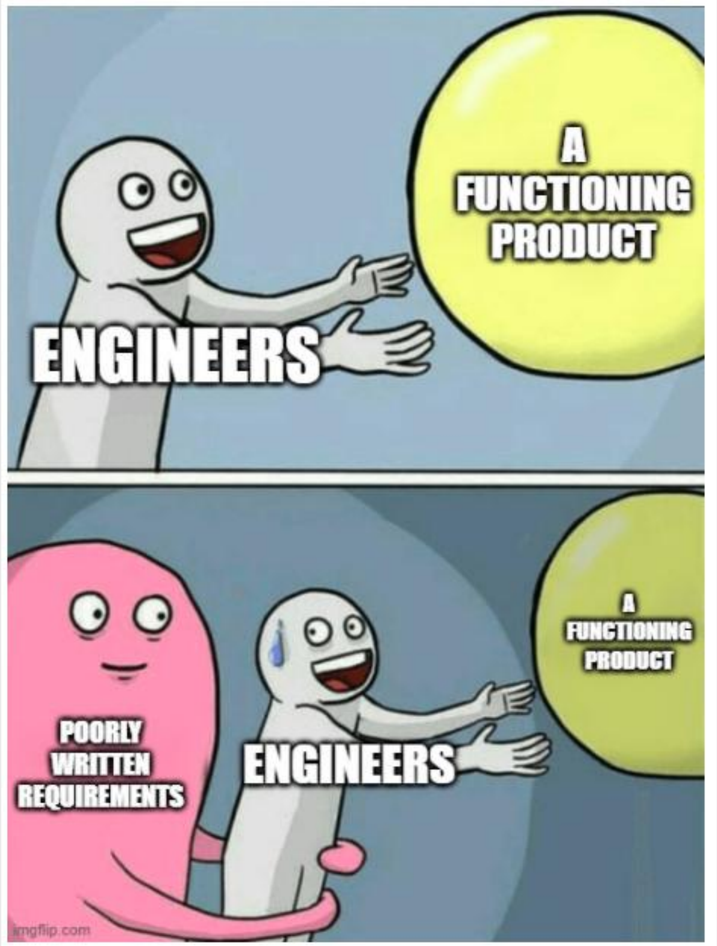
SESSION 06 – REQUIREMENT ENGINEERING

CONTENT OUTLINE

- Requirements engineering concepts
- Stakeholders
- Requirement types
- Elicitation techniques
- Analysis & validation
- Requirements monitoring & management

INTRODUCTION TO REQUIREMENTS ENGINEERING





INTRODUCTION TO REQUIREMENTS ENGINEERING

IMPORTANCE OF REQUIREMENTS ENGINEERING

- **Requirements Engineering (RE)** is the process of defining, documenting, and maintaining the requirements in the engineering design process.

Importance:

- **Foundation for Development**: Requirements serve as the foundation for all subsequent development activities.
- **Stakeholder Satisfaction**: Ensures the final product meets the needs and expectations of stakeholders.
- **Risk Reduction**: Identifies potential issues early, reducing the cost and time associated with rework.
- **Scope Management**: Helps in defining the project scope and prevents scope creep.

STAKEHOLDER IDENTIFICATION AND ANALYSIS

IDENTIFYING AND ANALYZING STAKEHOLDERS

- **Who are Stakeholders?**
  - Individuals or organizations affected by or can affect the system.
  - **Types**:
    - **Users**: Primary users, secondary users.
    - **Clients**: Those who commission the system.
    - **Developers**: Team members building the system.
    - **Regulators**: Authorities enforcing standards and regulations.

STAKEHOLDER IDENTIFICATION AND ANALYSIS

IDENTIFYING AND ANALYZING STAKEHOLDERS

- **Stakeholder Analysis**:
  - **Purpose**: Understand stakeholders' influence and interest.
  - **Steps**:
    - **Identify Stakeholders**: List all potential stakeholders.
    - **Assess Impact and Influence**: Determine their level of impact and influence.
    - **Prioritize Stakeholders**: Focus on key stakeholders.
    - **Understand Needs and Expectations**: Gather initial insights.

STAKEHOLDER IDENTIFICATION AND ANALYSIS

MANAGING STAKEHOLDER RELATIONSHIPS

- **Communication**: Establish clear channels and frequency.
- **Engagement**: Involve stakeholders throughout the project.
- **Expectation Management**: Align stakeholder expectations with project realities.
- **Conflict Resolution**: Address disagreements promptly and professionally.

TYPES OF REQUIREMENTS

FUNCTIONAL VS NON-FUNCTIONAL REQUIREMENTS

- **Functional Requirements:**
  - Define what the system should do.

  Examples:
  - “The system checks prerequisites before registration”
  - "The system shall allow users to create and edit profiles."
  - "The system shall generate monthly sales reports.”

- **Non-Functional Requirements**:
  - Define how the system performs its functions.
  - Include constraints and qualities.

  Categories: Performance, Security, Usability and Reliability

  Examples:
  - The system supports 10,000 concurrent users
  - The system shall lock a user account after 5 consecutive failed login attempts.
  - The system shall respond to user requests within 2 seconds for 95% of transactions under normal load.

TECHNIQUES FOR ELICITING REQUIREMENTS

OVERVIEW OF ELICITATION TECHNIQUES

- **Purpose**: Gather comprehensive requirements from stakeholders.
- **Challenges**:
  - **Ambiguity**: Unclear or incomplete information.
  - **Communication Barriers**: Misunderstandings due to jargon or assumptions.
  - **Stakeholder Availability**: Limited access to key stakeholders.

TECHNIQUES FOR ELICITING REQUIREMENTS

INTERVIEWS

- **Types**:
  - **Structured Interviews**: Predefined set of questions.
  - **Unstructured Interviews**: Open-ended conversations.
- **Best Practices**:
  - **Preparation**: Research and plan questions.
  - **Active Listening**: Pay attention and clarify doubts.
  - **Recording**: Take notes or record (with permission).

TECHNIQUES FOR ELICITING REQUIREMENTS

SURVEYS AND QUESTIONNAIRES

- **Use When**: Access to stakeholders is limited; need quantitative data.
- **Design Tips**:
  - **Clear Questions**: Avoid ambiguity.
  - **Logical Flow**: Organize questions coherently.
  - **Pilot Testing**: Test the questionnaire before distribution.

TECHNIQUES FOR ELICITING REQUIREMENTS

WORKSHOPS AND FOCUS GROUPS

- **Workshops**:
  - Collaborative sessions with stakeholders.
  - **Advantages**: Foster collaboration, immediate feedback.
- **Focus Groups**:
  - Guided discussions with selected stakeholders.
  - **Advantages**: Gain diverse perspectives.

TECHNIQUES FOR ELICITING REQUIREMENTS

OBSERVATION AND ETHNOGRAPHY

- **Observation**:
  - Watch stakeholders perform tasks.
  - **Benefits**: Understand actual workflows and pain points.
- **Ethnography**:
  - In-depth study of users in their environment.
  - **Benefits**: Gain deep insights into user behavior and culture.

DOCUMENTING STAKEHOLDER NEEDS AND EXPECTATIONS

WRITING CLEAR AND CONCISE REQUIREMENT SPECIFICATIONS

- **Guidelines**:
  - **Clarity**: Use simple, unambiguous language.
  - **Consistency**: Maintain uniform terminology.
  - **Completeness**: Cover all necessary aspects.
  - **Verifiability**: Requirements should be testable.

Example of a Well-Written Requirement:

- "The system shall authenticate users using a username and password before granting access to the dashboard."

ANALYZING AND VALIDATING REQUIREMENTS

REQUIREMENT ANALYSIS TECHNIQUES

- Prioritization:
  - **MoSCoW Method**: Must have, Should have, Could have, Won't have.
- Modeling:
  - **Data Flow Diagrams (DFDs)**: Represent data movement.
  - **Entity-Relationship Diagrams (ERDs)**: Show data relationships.

REQUIREMENTS TRACEABILITY AND MANAGEMENT

IMPORTANCE OF TRACEABILITY

- **Definition**: Ability to link requirements to their origins and track them throughout the project.
- **Benefits**:
  - **Impact Analysis**: Understand effects of changes.
  - **Verification**: Ensure all requirements are fulfilled.
  - **Accountability**: Maintain records of decisions.

REQUIREMENTS TRACEABILITY AND MANAGEMENT

TOOLS FOR REQUIREMENTS MANAGEMENT

- **Popular Tools**:
  - **IBM Rational DOORS**: For requirements management and traceability.
  - **Jama Software**: Collaborative requirements management.
  - **Atlassian JIRA**: With plugins for requirements tracking.
- **Features to Look For**:
  - **Traceability Matrices**: Visualize relationships between requirements and other artifacts.
  - **Collaboration Capabilities**: Support team communication.
  - **Integration**: Ability to work with other development tools.

NOW THAT WE KNOW THE TOOLS, LET’S SEE HOW REQUIREMENTS ENGINEERING WORK IS ORGANIZED CONCEPTUALLY.

REQUIREMENT ENGINEERING

Requirements engineering includes seven core tasks:

- Inception
- Elicitation
- Elaboration
- Negotiation
- Specification
- Validation
- Management

These tasks:

- **Do not occur strictly in sequence**
- Often **overlap and run in parallel**
- Are **adapted to each project**

REQUIREMENT ENGINEERING

Inception: Understanding the Problem

- Identify the **business need**
- Understand **who needs the system**
- Define the **problem boundaries**

Elicitation: Discovering Stakeholder Goals

- Engage stakeholders through discussion
- Identify **functional and non-functional goals**
- Capture **what success means**

Elaboration: Refining Requirements

- Refine user scenarios
- Clarify system behavior
- Identify key concepts and interactions

REQUIREMENT ENGINEERING

Negotiation: Resolving Conflicts

- Handle conflicting requirements
- Prioritize based on value, cost, and risk
- Reach agreements all parties can accept

Specification: Representing Requirements

- Requirements may be documented as:
  - Text
  - Models
  - Scenarios
  - Prototypes
- Level of formality depends on project size and risk

REQUIREMENT ENGINEERING

Validation: Checking Requirement Quality

- Ensure requirements are:
  - Clear
  - Consistent
  - Testable
  - Achievable
- Detect errors, omissions, and ambiguity

Requirements Management: Handling Change

- Track requirement changes
- Maintain traceability
- Control scope while accepting evolution

SINGLE CASE STUDY: UNIVERSITY ONLINE COURSE REGISTRATION SYSTEM

Case: Online Course Registration System

Problem:

- Students struggle to register courses on time
- Manual approvals cause delays
- Errors lead to schedule conflicts

Goal:

- Improve student experience
- Reduce administrative workload

SINGLE CASE STUDY: UNIVERSITY ONLINE COURSE REGISTRATION SYSTEM

Inception: Understanding the Problem

- Business need: faster, error-free registration
- Stakeholders:
  - Students
  - Academic advisors
  - Registrar office
- Scope:
  - Course selection
  - Schedule validation
  - Approval workflow

SINGLE CASE STUDY: UNIVERSITY ONLINE COURSE REGISTRATION SYSTEM

Elicitation: Discovering Stakeholder Goals

Stakeholder goals:

- Students: register quickly, avoid clashes
- Advisors: approve efficiently
- Admins: enforce academic rules

Non-functional goals:

- Availability during peak time
- Security of student data

SINGLE CASE STUDY: UNIVERSITY ONLINE COURSE REGISTRATION SYSTEM

Elaboration: Refining Requirements

User scenarios refined:

- Student selects courses
- System checks prerequisites
- System detects schedule conflicts
- Advisor reviews exceptions

Key concepts identified:

- Course
- Student
- Prerequisite
- Schedule

SINGLE CASE STUDY: UNIVERSITY ONLINE COURSE REGISTRATION SYSTEM

Negotiation: Resolving Conflicts

Conflicts:

- Students want unlimited course selection
- University enforces credit limits

Resolution:

- System allows requests beyond limit
- Advisor approval required

Specification: Representing Requirements

Requirements represented as:

- User stories (student actions)
- Acceptance criteria
- Simple workflow diagrams

Formality:

- Lightweight documentation
- Detailed rules for validation logic

SINGLE CASE STUDY: UNIVERSITY ONLINE COURSE REGISTRATION SYSTEM

Validation: Checking Requirement Quality

Validation checks:

- “Fast registration” clarified as: Page loads < 2 seconds
- “User-friendly” replaced with: Max 3 steps to register

Stakeholders review and confirm requirements.

Requirements Management: Handling Change

Change examples:

- New prerequisite policy introduced
- Course capacity rules updated

Management actions:

- Update backlogTrace affected requirements
- Revalidate impacted features

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To replace programmers with AI, clients will need to accurately describe what they want.

We're safe.

CASE STUDY AND INTERACTIVE EXERCISE

CASE STUDY: ONLINE RETAIL SYSTEM

Scenario:

A company wants to develop an online retail platform that allows customers to browse products, place orders, and track shipments.

Tasks:

- **Part 1**: Identify key stakeholders.
- **Part 2**: Choose appropriate elicitation techniques for gathering requirements from these stakeholders.
- **Part 3**: Write one functional and one non-functional requirement based on the scenario.
- **Part 4**: Using the Online Retail System, briefly describe how requirements evolve across the seven RE tasks

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